

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$D = 5 + 2 \sin(30 \times 6.5)^\circ = \text{awrt } 4.48 \text{ m with units}$	B1	3.4
		(1)	
(b)	$3.8 = 5 + 2 \sin(30t)^\circ \Rightarrow \sin(30t)^\circ = -0.6$	M1 A1	1.1b 1.1b
	$t = 10.77$	dM1	3.1a
	10:46 a.m. or 10:47 a.m.	A1	3.2a
		(4)	
(5 marks)			

Notes:

(a)

B1: Scored for using the model ie. substituting $t = 6.5$ into $D = 5 + 2 \sin(30t)^\circ$ and stating $D = \text{awrt } 4.48 \text{ m}$. The units must be seen somewhere in (a). So allow when $D = 4.482... = 4.5 \text{ m}$
 Allow the mark for a correct answer without any working.

(b)

M1: For using $D = 3.8$ and proceeding to $\sin(30t)^\circ = k$, $|k| \leq 1$

A1: $\sin(30t)^\circ = -0.6$ This may be implied by any correct answer for t such as $t = 7.2$

If the A1 implied, the calculation must be performed in degrees.

dM1: For finding the first value of t for their $\sin(30t)^\circ = k$ after $t = 8.5$.

You may well see other values as well which is not an issue for this dM mark

(Note that $\sin(30t)^\circ = -0.6 \Rightarrow 30t = 216.9^\circ$ as well but this gives $t = 7.2$)

For the correct $\sin(30t)^\circ = -0.6 \Rightarrow 30t = 323.1 \Rightarrow t = \text{awrt } 10.8$

For the incorrect $\sin(30t)^\circ = +0.6 \Rightarrow 30t = 396.9 \Rightarrow t = \text{awrt } 13.2$

So award this mark if you see $30t = \text{inv sin their } -0.6$ to give the first value of t where $30t > 255$

A1: Allow 10:46 a.m. (12 hour clock notation) or 10:46 (24 hour clock notation) oe
 Allow 10:47 a.m. (12 hour clock notation) or 10:47 (24 hour clock notation) oe
 DO NOT allow 646 minutes or 10 hours 46 minutes.

Q2.

Question	Scheme	Marks	AOs
(a)	$\cos 3A = \cos(2A + A) = \cos 2A \cos A - \sin 2A \sin A$	M1	3.1a
	$= (2 \cos^2 A - 1) \cos A - (2 \sin A \cos A) \sin A$	dM1	1.1b
	$= (2 \cos^2 A - 1) \cos A - 2 \cos A (1 - \cos^2 A)$	ddM1	2.1

	$= 4\cos^3 A - 3\cos A^*$	A1*	1.1b
		(4)	
(b)	$1 - \cos 3x = \sin^2 x \Rightarrow \cos^2 x + 3\cos x - 4\cos^3 x = 0$	M1	1.1b
	$\Rightarrow \cos x(4\cos^2 x - \cos x - 3) = 0$ $\Rightarrow \cos x(4\cos x + 3)(\cos x - 1) = 0$ $\Rightarrow \cos x = \dots$	dM1	3.1a
	Two of $-90^\circ, 0, 90^\circ$, awrt 139°	A1	1.1b
	All four of $-90^\circ, 0, 90^\circ$, awrt 139°	A1	2.1
		(4)	
(8 marks)			

Notes:

(a)

Allow a proof in terms of x rather than A

M1: Attempts to use the compound angle formula for $\cos(2A + A)$ or $\cos(A + 2A)$

Condone a slip in sign

dM1: Uses correct double angle identities for $\cos 2A$ and $\sin 2A$

$\cos 2A = 2\cos^2 A - 1$ must be used. If either of the other two versions are used expect to see an attempt to replace $\sin^2 A$ by $1 - \cos^2 A$ at a later stage.

Depends on previous mark.

ddM1: Attempts to get all terms in terms of $\cos A$ using correct and appropriate identities.

Depends on both previous marks.

A1*: A completely correct and rigorous proof including correct notation, no mixed variables, missing brackets etc.

Alternative right to left is possible:

$$4\cos^3 A - 3\cos A = \cos A(4\cos^2 A - 3) = \cos A(2\cos^2 A - 1 + 2(1 - \sin^2 A) - 2) = \cos A(\cos 2A - 2\sin^2 A)$$

$$= \cos A \cos 2A - 2\sin A \cos A \sin A = \cos A \cos 2A - \sin 2A \sin A = \cos(2A + A) = \cos 3A$$

Score M1: For $4\cos^3 A - 3\cos A = \cos A(4\cos^2 A - 3)$

dM1: For $\cos A(2\cos^2 A - 1 + 2(1 - \sin^2 A) - 2)$ (Replaces $4\cos^2 A - 1$ by $2\cos^2 A - 1$ and $2(1 - \sin^2 A)$)

ddM1: Reaches $\cos A \cos 2A - \sin 2A \sin A$

A1: $\cos(2A + A) = \cos 3A$

(b)

M1: For an attempt to produce an equation just in $\cos x$ using both part (a) and the identity $\sin^2 x = 1 - \cos^2 x$

Allow one slip in sign or coefficient when copying the result from part (a)

dM1: Dependent upon the preceding mark. It is for taking the cubic equation in $\cos x$ and making a valid attempt to solve. This could include factorisation or division of a $\cos x$ term followed by an attempt to solve the 3 term quadratic equation in $\cos x$ to reach at least one non zero value for $\cos x$.

May also be scored for solving the cubic equation in $\cos x$ to reach at least one non zero value for $\cos x$.

A1: Two of $-90^\circ, 0, 90^\circ$, awrt 139° Depends on the first method mark.

A1: All four of $-90^\circ, 0, 90^\circ$, awrt 139° with no extra solutions offered within the range.

Note that this is an alternative approach for obtaining the cubic equation in (b):

$$1 - \cos 3x = \sin^2 x \Rightarrow 1 - \cos 3x = \frac{1}{2}(1 - \cos 2x)$$

$$\Rightarrow 2 - 2\cos 3x = 1 - \cos 2x$$

$$\Rightarrow 1 = 2\cos 3x - \cos 2x$$

$$\Rightarrow 1 = 2(4\cos^3 x - 3\cos x) - (2\cos^2 x - 1)$$

$$\Rightarrow 0 = 4\cos^3 x - 3\cos x - \cos^2 x$$

The M1 will be scored on the penultimate line when they use part (a) and use the correct identity for $\cos 2x$

Q3.

Question	Scheme	Marks	AOs
(a)	$R = \sqrt{109}$	B1	1.1b
	$\tan \alpha = \frac{3}{10}$	M1	1.1b
	$\alpha = 16.70^\circ$ so $\sqrt{109} \cos(\theta + 16.70^\circ)$	A1	1.1b
		(3)	
(b)	(i) e.g. $H = 11 - 10 \cos(80t)^\circ + 3 \sin(80t)^\circ$ or $H = 11 - \sqrt{109} \cos(80t + 16.70)^\circ$	B1	3.3
	(ii) $11 + \sqrt{109}$ or 21.44 m	B1ft	3.4
		(2)	
(c)	Sets $80t + "16.70" = 540$	M1	3.4
	$t = \frac{540 - "16.70"}{80} = (6.54)$	M1	1.1b
	$t = 6$ mins 32 seconds	A1	1.1b
		(3)	
(d)	Increase the '80' in the formula For example use $H = 11 - 10 \cos(90t)^\circ + 3 \sin(90t)^\circ$		3.3
		(1)	
(9 marks)			

Notes:
(a) B1: $R = \sqrt{109}$ Do not allow decimal equivalents M1: Allow for $\tan \alpha = \pm \frac{3}{10}$ A1: $\alpha = 16.70^\circ$
(b)(i) B1: see scheme (b)(ii) B1ft: their $11 +$ their $\sqrt{109}$ Allow decimals here.
(c) M1: Sets $80t + "16.70" = 540$. Follow through on their 16.70 M1: Solves their $80t + "16.70" = 540$ correctly to find t A1: $t = 6$ mins 32 seconds
(d) B1: States that to increase the speed of the wheel the 80's in the equation would need to be increased.

Q4.

Question	Scheme	Marks	AOs
(a)(i)	$y \times \frac{dx}{dt} = 5 \sin 2t \times 6 \cos t$ or $5 \times 2 \sin t \cos t \times 6 \cos t$	M1	1.2
	$(\text{Area} =) \int 5 \sin 2t \times 6 \cos t \, dt = \int 5 \times 2 \sin t \cos t \times 6 \cos t \, dt$ or $\int 5 \sin 2t \times 6 \cos t \, dt = \int 60 \sin t \cos^2 t \, dt$	dM1	1.1b
	$(\text{Area} =) \int_0^{\frac{\pi}{2}} 60 \sin t \cos^2 t \, dt *$	A1*	2.1*
	(3)		
(a)(ii)	$\int 60 \sin t \cos^2 t \, dt = -20 \cos^3 t$	M1 A1	1.1b 1.1b
	$\text{Area} = \left[-20 \cos^3 t \right]_0^{\frac{\pi}{2}} = 0 - (-20) = 20 *$	A1*	2.1
	(3)		
(b)	$5 \sin 2t = 4.2 \Rightarrow \sin 2t = \frac{4.2}{5}$	M1	3.4
	$t = 0.4986..., 1.072...$	A1	1.1b
	Attempts to find the x values at both t values	dM1	3.4
	$t = 0.4986... \Rightarrow x = 2.869...$ $t = 1.072 \Rightarrow x = 5.269...$	A1	1.1b
	Width of path = 2.40 metres	A1	3.2a
	(5)		
(11 marks)			

Notes:

(a)(i)

M1: Attempts to multiply y by $\frac{dx}{dt}$ to obtain $A \sin 2t \cos t$ but may apply $\sin 2t = 2 \sin t \cos t$ here

dM1: Attempts to use $\sin 2t = 2 \sin t \cos t$ within an integral which may be implied by

$$\text{e.g. } \int \sin 2t \times \cos t \, dt = \int k \sin t \cos^2 t \, dt$$

A1*: Fully correct work leading to the given answer.

This must include $\sin 2t = 2 \sin t \cos t$ or e.g. $5 \sin 2t = 10 \sin t \cos t$ seen explicitly in their proof and a correct intermediate line that includes an integral sign and the “ dt ”

Allow the limits to just “appear” in the final answer e.g. working need not be shown for the limits.

(a)(ii)

M1: Obtains $\int 60 \sin t \cos^2 t \, dt = k \cos^3 t$. This may be attempted via a substitution of $u = \cos t$ to obtain

$$\int 60 \sin t \cos^2 t \, dt = k u^3$$

A1: Correct integration $-20 \cos^3 t$ or equivalent e.g. $-20u^3$

A1*: Rigorous proof with all aspects correct including the correct limits and the $0 - (-20)$ and

$$\text{not just: } -20 \cos^3 \frac{\pi}{2} - (-20 \cos^3 0) = 20$$

(b)

M1: Uses the given model and attempts to find value(s) of t when $\sin 2t = \frac{4.2}{5}$. Look for $2t = \sin^{-1} \frac{4.2}{5} \Rightarrow t = ...$

A1: At least one correct value for t , correct to 2 dp. FYI $t = 0.4986..., 1.072...$ or in degrees $t = 28.57..., 61.42...$

dM1: Attempts to find **TWO** distinct values of x when $\sin 2t = \frac{4.2}{5}$. Condone poor trig work and allow this mark if 2

values of x are attempted from 2 values of t .

A1: Both values correct to 2 dp. NB $x = 2.869\dots, 5.269\dots$

Or may take Cartesian approach

$$5 \sin 2t = 4.2 \Rightarrow 10 \sin t \cos t = 4.2 \Rightarrow 10 \frac{x}{6} \sqrt{1 - \frac{x^2}{36}} = 4.2 \Rightarrow x^4 - 36x^2 + 228.6144 = 0 \Rightarrow x = 2.869\dots, 5.269\dots$$

M1: For converting to Cartesian form A1: Correct quartic M1: Solves quartic A1: Correct values

A1: 2.40 metres or 240 cm

Allow awrt 2.40 m or allow 2.4m (not awrt 2.4 m) and allow awrt 240 cm. Units are required.

Q5.

Question	Scheme	Marks	AOs
(a)	$R = \sqrt{5}$	B1	1.1b
	$\tan \alpha = 2 \Rightarrow \alpha = \dots$	M1	1.1b
	$\alpha = 1.107$	A1	1.1b
		(3)	
	$\theta = 5 + \sqrt{5} \sin\left(\frac{\pi t}{12} + 1.107 - 3\right)$		
(b)	$(5 + \sqrt{5})^\circ\text{C}$ or awrt 7.24°C	B1ft	2.2a
		(1)	
(c)	$\frac{\pi t}{12} + 1.107 - 3 = \frac{\pi}{2} \Rightarrow t =$	M1	3.1b
	$t =$ awrt 13.2	A1	1.1b
	Either 13:14 or 1:14 pm or 13 hours 14 minutes after midnight.	A1	3.2a
		(3)	
(7 marks)			
Notes:			



(a)

B1: $R = \sqrt{5}$ only.**M1:** Proceeds to a value of α from $\tan \alpha = \pm 2$, $\tan \alpha = \pm \frac{1}{2}$, $\sin \alpha = \pm \frac{2}{\sqrt{5}}$ OR $\cos \alpha = \pm \frac{1}{\sqrt{5}}$

It is implied by either awrt 1.11 (radians) or 63.4 (degrees)

A1: $\alpha =$ awrt 1.107

(b)

B1ft: Deduces that the maximum temperature is $(5 + \sqrt{5})^\circ\text{C}$ or awrt 7.24°C Remember to isw
Condone a lack of units. Follow through on their value of R so allow $(5 + R)^\circ\text{C}$

(c)

M1: An complete strategy to find t from $\frac{\pi t}{12} \pm 1.107 - 3 = \frac{\pi}{2}$.

Follow through on their 1.107 but the angle must be in radians.

It is possible via degrees but only using $15t \pm 63.4 - 171.9 = 90$ **A1:** awrt $t = 13.2$ **A1:** The question asks for the time of day so accept either 13:14, 1:14 pm, 13 hours 14 minutes after midnight, 13h 14, or 1 hour 14 minutes after midday. If in doubt use review

It is possible to attempt parts (b) and (c) via differentiation but it is unlikely to yield correct results.

$$\frac{d\theta}{dt} = \frac{\pi}{12} \cos\left(\frac{\pi t}{12} - 3\right) - \frac{2\pi}{12} \sin\left(\frac{\pi t}{12} - 3\right) = 0 \Rightarrow \tan\left(\frac{\pi t}{12} - 3\right) = \frac{1}{2} \Rightarrow t = 13.23 = 13:14 \text{ scores M1 A1 A1}$$

$$\frac{d\theta}{dt} = \cos\left(\frac{\pi t}{12} - 3\right) - 2 \sin\left(\frac{\pi t}{12} - 3\right) = 0 \Rightarrow \tan\left(\frac{\pi t}{12} - 3\right) = \frac{1}{2} \Rightarrow t = 13.23 = 13:14 \text{ they can score M1 A0 A1 (SC)}$$

A value of $t = 1.23$ implies the minimum value has been found and therefore incorrect method M0.